

ZS-F43

The Charge-Unit Gauge Principle: Representation-Theoretic Unit-Invisibility, One-Residual Uniformization of the B3 Charge Arc, and the Two-Ledger Conditional Charge Unit ($\hat{e}_6 = 2\pi e^{-4\pi Q}$)

Kenny Kang — Z-Spin Cosmology Program (independent) July 2026 · Foundations Series · Paper code ZS-F43 · Version 1.1 Re-typing terminus of the charge-unit arc ZS-F33 $\rightarrow$ ZS-F34 $\rightarrow$ ZS-F35 $\rightarrow$ ZS-F36. Consumes ZS-F32, ZS-F38–F42, ZS-A25, ZS-A31, ZS-A32, ZS-M46/M47.

Verification: 40/40 PASS + 10/10 guards (`zs_f43_verify_v1_1.py`; fail-closed, exits non-zero on any failure) | 7 firewalled observations printed separately, never counted as PASS | **Zero fitted parameters** | $(\mathbf{A}, \mathbf{Q}, \dim \mathbf{Z}) = (35/437, 11, 2)$ LOCKED.

§0. Abstract

ZS-F42 v1.9 terminated the B3 charge arc at a terminal NON-CLAIM: the dimensionful membrane charge (e_6) behind the ZS-F33 Charge-Unit Obstruction is a genuine OPEN that $(\mathbf{A}, \mathbf{Q})$ and topology provably cannot fix. This paper does not close that OPEN. It re-types it, in four theorems.

(T1) *Unit-Invisibility* [general theorem IMPORTED-PROVEN; Z-Spin instance DERIVED-CONDITIONAL]. In the compact odd three-form zero-mode sector (ZS-F32.24–26) with the unimodular pair $(\hat{\Lambda}, \hat{T}_4 = i\hbar)$ (ZS-A25), the observable algebra is generated by the dimensionless Weyl pair $(\hat{n}, \hat{\varphi})$ on $(\mathbb{Z} \times U(1))$ with $(\hat{n}, \hat{\varphi}) = i$. By the Stone–von Neumann theorem in Mackey's locally-compact form, its irreducible representation is unique up to unitary equivalence, and the charge unit (e_-^2) enters only through the spectral dictionary $(\Lambda = e_-^2(\hat{n} + \omega/2\pi))$. Hence (e_6) is not an observable of the Z-sector algebra alone; it becomes physical only through the gravitational pairing, and the well-posed residual of barrier B3 is the dimensionless $(\hat{e}_6 := e_6/\bar{M}_P^2)$, never dimensionful (e_6). This paper adds the algebraic form of the statement (Theorem F43.T1'): the charge unit is not an invariant of the compact zero-mode Weyl algebra, so its previous terminal status is re-typed into a **conditional, sector-internal no-go theorem** — under R1–R3, the Z-sector algebra alone cannot fix a dimensionful charge unit.

(T2) *One-Residual Uniformization* [DERIVED as bookkeeping; value leg conditional]. Modulo the two dimensionless conditional dressings $(\alpha_{\rm UV})$ (ZS-F36) and $(C_{\rm UV})$ (OPEN), the four residual names of the arc — (e_6), $(M_{\rm UV})$, (M_K) , and the metric-scale gate — are one object, the Koenigs linearizing length in Planck units $(\hat{\ell} = \ell/\bar{\ell}_P)$, with $(\hat{e}_6 = (2\pi/\alpha_{\rm UV})\hat{\ell}^2)$. The ZS-F39 modular length functor localizes it as $(\hat{e}_6 = 2\pi(\bar{\ell}_P/E_{\rm len})^2)$, identifying the charge arc's OPEN with the ZS-M47 Parent-Factor Realization OPEN. At the ZS-F38 depth ($t^* = Q$) the conditional value is $(\hat{e}_6 = 2\pi e^{-4\pi Q}) \approx 5.83$

$\times 10^{-60}$), i.e. $(\sqrt{e_6} = \sqrt{2\pi}, \bar{M}_P e^{-2\pi Q} \approx 5.88) \text{ meV}$ (canonical normalization); the value inherits the ZS-A32 statistical status (HYPOTHESIS-strong, executed pre-registered MC, single-target ($p = 0.50\%$)) and performs no mechanism promotion (ZS-F40 terminal verdict respected).

(T3) *Arithmetic-Leg Rigidity* [IMPORTED-PROVEN + DERIVED-CONDITIONAL]. Under exact large-gauge invariance the integer ledger of the arc — Smith normal form 1, the dimensionless Wess–Zumino phase (2π), the flux-step counting — is 1PI-exact; provided the parent theory carries no additional metric-side threshold operator, the field-redefinition-invariant unit flows only through the single metric stiffness (C_{UV}). The entire quantum debt of the ZS-F35/F36 line is thereby localized in one Kähler coefficient.

(T4) *Two-Ledger Consistency* [form DERIVED-CONDITIONAL; value REGRESSION, firewalled]. The UV ledger (T2) and the IR frozen-gas ledger (ZS-F42.7/9) evaluate the same $(\hat{e}_6)$; their consistency is the single dimensionless identity $(C_{\text{UV}} = c_{\chi}, (H_{\text{partial}}/\bar{M}_P)^2, e^{8\pi Q}, (1260/4807))$, which — behind the derivation $\perp$ regression firewall — evaluates to $(C_{\text{UV}} = 1.244)$, inside the ZS-A32 band (≈ 1.24), reproducing the ZS-A31 regression ($M_{\text{eff}} = 2.48) \text{ meV}$. T4 is definitionally equivalent to ZS-A31/A32 re-expressed in charge-unit language: no new empirical content. The new content is the binding of two conditional chains to one named object and one pre-registered joint falsification gate.

Barrier B3 is not closed. What changes is its type: the terminal NON-CLAIM is re-typed into a conditional sector-internal no-go theorem (nothing internal to the sector could ever have fixed (e_6)); four residual names become one; the quantum debt becomes one coefficient; and the single remaining dimensionless unknown $(\hat{e}_6)$ carries two agreeing conditional evaluations and a shared-OPEN localization at (E_{len}) . Verification: 40/40 PASS + 10/10 guards; zero fitted parameters; $(\mathbf{A}, \mathbf{Q}, \dim \mathbf{Z}) = (35/437, 11, 2)$ LOCKED.

Epistemic Status Legend

Tag	Meaning in this paper
PROVEN	Established within the corpus by exact mathematics (often exact rational arithmetic), machine-verified.
IMPORTED-PROVEN	External theorem with published proof, consumed as-is (Stone–von Neumann–Mackey; Dirac quantization; Hopkins–Singer discreteness). Never claimed as a corpus result.
DERIVED	Follows from corpus axioms/locked constants by explicit steps, no additional hypotheses.

Tag	Meaning in this paper
DERIVED-BY-INHERITANCE	Verbatim consumption of an upstream corpus result at its stated version and status.
DERIVED-CONDITIONAL	Derived contingent on explicitly listed hypotheses or open registries (conditions named at point of use).
HYPOTHESIS-strong	Survives a pre-registered anti-numerology Monte Carlo (here: ZS-A32, single-target ($p = 0.50\%$)); not a derivation.
OBSERVATION	Conceptual or empirical anchor used for orientation; carries no proof burden.
REGRESSION (FIREWALLED)	Comparison against external observational data behind the derivation $\perp$ regression firewall; printed separately; never counted as PASS.
IDENTITY	True by construction/definition; recorded to prevent it being mistaken for evidence.
NON-CLAIM	Explicit registration that a statement is not being made.
OPEN	Well-posed and unresolved.
NO-GO	Proven impossibility under stated conditions.

§1. Introduction: the inherited terminus and the re-typing mandate

The charge-unit arc of the Foundations series ended, honestly, three times. ZS-F35 v1.5 proved the structural factor ($36\mathbf{A}/\mathbf{Q} = 1260/4807$) and the multiplicity-free duality-singlet ($(\nu_s^2 = 6 = \dim Y)$), and registered the normalization gate and the residual scale ($M_{\rm UV}$) as what remained. ZS-F36 v2.1 proved the integer ledger — Smith normal form 1 of the wrapped pairing and the dimensionless Wess–Zumino phase (2π) — reduced the normalization gate to ($c_e = 2\pi \rightarrow \alpha_{\rm UV} = 1$) (DERIVED-CONDITIONAL), relocated ($M_{\rm UV}$) onto the parent scale (M_K) by the radion argument, proved ($M_{\rm UV}$) irreducible as a corpus-internal derivation target, and left two gates OPEN: the full parent one-particle-irreducible stiffness ($C_{\rm UV}$) and the metric-scale gate ($M_K/\bar{M}_P$). ZS-F42 v1.9 then closed the arc's accounting: the susceptibility ($\chi_- = e_-^2/(4\pi^2 Z_-)$) is a dimensionful quantity which (**A**, **Q**) plus topology provably cannot fix (the ZS-F33.8 Charge-Unit Obstruction), the residual is the membrane charge (e_6), and the terminus was registered as a NON-CLAIM with a prohibition on direct-derivation attempts (gate F-F42.36).

In parallel, ZS-F40 v1.2 recorded the determinant/clock mechanism programme for the depth factor ($e^{-2\pi Q}$) as TERMINAL after a five-route audit (reopening only through gate F-F40.6), and ZS-F41

v1.1 installed a Stopping Rule against speculative mechanism extensions. Any further paper on this arc must therefore add value without doing the two forbidden things: deriving $(\hat{e}_6)$ directly, and promoting the depth mechanism.

The present paper starts from an observation about the corpus's own honest statements. Every terminal statement the arc produced about its residual is a statement about the residual's *type*, not its value: ZS-F33.8 ("flux integrality fixes the number of quanta, never the unit"); ZS-A25 ("exactly one dimensionful datum enters the sector"); ZS-A31 ("the forbidden absolute scale versus the merely-open ratio"); ZS-F42.34 ("external mathematics confirms the obstruction, it does not lift it"). This paper takes the hint literally and proves the type statement. Concretely:

(i) Theorem T1 proves that the charge unit is *invisible from inside* the Z-sector zero-mode algebra — a representation-theoretic fact (Stone–von Neumann–Mackey), not a toolkit limitation — so the well-posed unknown of barrier B3 is the dimensionless $(\hat{e}_6 := e_6 / \bar{M}_P^2)$, never dimensionful (e_6) . Its algebraic form (Theorem F43.T1') states this as a property of the compact zero-mode Weyl algebra, so that the result reads as an instance of a proven general theorem rather than a Z-Spin-internal assertion. (ii) Theorem T2 shows that, modulo the two dimensionless conditional dressings already registered upstream, the arc's four residual names are one modulus, localized by the ZS-F39 length functor at the seam; the charge frontier and the ZS-M47 Parent-Factor Realization frontier are the same OPEN. (iii) Theorem T3 localizes the entire quantum debt of the arc in the single coefficient $(C_{\rm UV})$. (iv) Theorem T4 binds the two existing conditional evaluations of $(\hat{e}_6)$ — the UV depth reading and the IR frozen-gas reading — into one consistency identity, evaluated only behind the firewall.

Equally important is what this paper does not do. It does not close B3: $(\hat{e}_6)$ remains OPEN as a derivation. It does not touch the ZS-F40 terminal verdict: the value leg of T2 inherits HYPOTHESIS-strong from ZS-A32 and nothing more. And T4 contains no new empirical content: as §6.4 states explicitly, it is ZS-A31/A32 re-expressed; its worth is conceptual binding plus a pre-registered joint falsification gate. The paper's product is a well-posed target with a theorem explaining why nothing less abstract could have worked.

Notation and conventions. $(\hat{e}_6 := e_6 / \bar{M}_P^2)$ and $(\hat{\ell} := \ell / \bar{\ell}_P)$ (reduced Planck units); $(H_{\rm partial})$ denotes the present-epoch horizon rate consumed as an IR datum where explicitly firewalled; statuses follow the Legend above and are attached at point of use. Terminology follows the corpus convention: *Z-sector* names the stage ($\dim \mathbf{Z} = 2$), *Z-Spin* names the action on it; sector-tagged structural terms (Z-channel, Z-anchor) are unchanged. Dimensionful exponentials are written in explicit LaTeX form throughout $((e^{-4\pi Q})$, $(e^{8\pi Q})$, $(C_{\rm UV}^{1/4})$, $(\hat{\ell}^{-2})$, $(M_{\rm UV}^{-1})$) to remove any plaintext-rendering ambiguity. All numerical claims in the body are reproduced by the fail-closed verification script `zs_f43_verify_v1_1.py` (40/40 PASS + 10/10 guards); every scale-bearing number appears only in firewalled context.

§2. Setup: the compact odd three-form zero mode as a Weyl pair

2.1 The corpus pair

ZS-F32.24 promotes the odd three-form (F_-) to a compact ($U(1)$) three-form gauge field: fluxes are Dirac-quantized, large gauge transformations act, and the zero-mode phase space is the cylinder rather than the plane. ZS-F32.25–26 selects the ($k^* = 0$) branch and derives the dimensionless branch spectrum $((2\pi k + \omega)^2)$, ($k \in \mathbb{Z}$), where the holonomy offset is corpus-locked seam data, ($\theta_Z = \omega = \arg \lambda^* = 2.2592495540$) (reproduced from the i-tetration fixed point (z^*) in verification Block K). ZS-A25 supplies the conjugate pair of unimodular-type gravity, ($[\hat{\Lambda}, \hat{T}_4] = i\hbar$) (Henneaux–Teitelboim; Unruh), together with its no-go: exactly one dimensionful datum enters the sector through this pair.

2.2 The external universality class

The construction sits in a proven external class. Brown–Teitelboim showed that membrane nucleation steps the effective cosmological term by amounts set by the membrane coupling; Bousso–Polchinski showed that with quantized fluxes the cosmological term moves on a lattice whose spacing is set by the fundamental charge (q^2). In both, the theory fixes the integers while the unit (q^2) is an input from outside the flux sector. The Z-Spin corpus belongs to this class with one extra discipline: its integers are corpus-derived and PROVEN (Smith normal form 1; Wess–Zumino phase (2π); ZS-F36), so the entire input question is concentrated in the unit — which is exactly the object this paper re-types.

2.3 The dimensionless Weyl pair

Define, on the ($k^* = 0$) branch with offset absorbed,

$[\hat{n} := (\hat{\Lambda} - \Lambda_0)/e_-^2, \quad \hat{\varphi} := e_-^2 \hat{T}_4 / \hbar, \quad \text{so that} \quad [\hat{n}, \hat{\varphi}] = i.]$

Large-gauge compactness makes ($\hat{\varphi}$) an angle ($(\hat{\varphi} \in U(1))$) and the spectrum of ($\hat{n}$) integer; the pair ($(\hat{n}, \hat{\varphi})$) lives on ($\mathbb{Z} \times U(1)$). The commutator computation is machine-checked symbolically (verification B1): the charge unit (e_-^2) cancels identically, entering the description only through the spectral dictionary ($\Lambda = e_-^2(\hat{n} + \omega/2\pi)$). The dimensionless branch spectrum $((2\pi k + \omega)^2)$ carries no (e_-^2) (verification B6). This is the whole setup; Theorem T1 is the statement that this cancellation is not an artifact of coordinates but a property of the observable algebra.

§3. Theorem F43.T1 — Unit-Invisibility

3.1 Statement

Let ($\mathcal{A}_Z$) denote the zero-mode observable algebra of the compact odd three-form sector: the von Neumann algebra generated by the Weyl unitaries of the pair ($(\hat{n}, \hat{\varphi})$) on ($\mathbb{Z} \times U(1)$). Assume:

(R1) the compact- $(U(1))$ promotion and branch selection of ZS-F32.24–26, inherited with their stated conditions; (R2) exact large-gauge invariance; (R3) the only dimension-bearing coupling out of the sector is the gravitational pairing of (Λ) into the Friedmann sector.

Then: (i) [DERIVED] $(\mathcal{A}_Z)$ is independent of (e_-^2) : the defining Weyl relations contain no occurrence of the charge unit, which enters only through the spectral dictionary $(\Lambda = e_-^2(\hat{n} + \omega/2\pi))$. (ii) [IMPORTED-PROVEN] By the Stone–von Neumann theorem in Mackey's locally-compact form applied to $(G = \mathbb{Z})$ (with dual $(\hat{G} = U(1))$), the irreducible representation of the associated Heisenberg system is unique up to unitary equivalence; hence no superselection datum inside the sector distinguishes two values of (e_-^2) . (iii) [DERIVED-CONDITIONAL on R1–R3] Therefore (e_6) — equivalently (e_-^2) , up to the PROVEN lattice norm $(\nu_s^2 = 6)$ of ZS-F35 — is not an observable of $(\mathcal{A}_Z)$; every observable functional of the charge unit factors through a dimensionless combination with an external scale, and under (R3) the minimal such combination is $(\hat{e}_6 := e_6/\bar{M}_P^2)$.

3.2 Proof

Step 1 (algebraic independence; DERIVED). The Weyl relations of $(\hat{n}, \hat{\varphi})$ on $(\mathbb{Z} \times U(1))$ — $(W(\theta) V(m) = e^{i m \theta} V(m) W(\theta))$, $(m \in \mathbb{Z}, \theta \in U(1))$ — are written entirely in dimensionless data; (e_-^2) appears in no relation. Machine check: the symbolic commutator $([\Lambda/e_-^2, e_-^2 \hat{T}_4/\hbar] = i)$ with (e_-^2) absent from the result (B1); the finite clock-shift model $(WS = \zeta SW)$ with $(\zeta = e^{2\pi i/N})$ contains no charge unit, and the generators built "at two different values of (e_-^2) " are literally the same operators (B4, B7).

Step 2 (uniqueness; IMPORTED-PROVEN). Stone–von Neumann, in the locally-compact-abelian form proved by Mackey [1–3], states that the irreducible covariant pair over $(G \times \hat{G})$ is unique up to unitary equivalence. For $(G = \mathbb{Z})$ this is the uniqueness of the clock-shift system on $(\ell^2(\mathbb{Z}))$. Consequently there is no representation-theoretic label — no superselection sector, no central charge, no inequivalent quantization — inside $(\mathcal{A}_Z)$ that could encode the value of (e_-^2) . The dictionary is the only door, and the dictionary is not an observable of the algebra.

Step 3 (egress; DERIVED-CONDITIONAL). An observable consequence of (e_-^2) must therefore pair sector data with something outside $(\mathcal{A}_Z)$. Under (R3) the only dimension-bearing egress is gravitational: (Λ) enters the Friedmann sector, whose scale is set by $(\bar{M}_P)$. The minimal invariant combination is the dimensionless ratio $(\hat{e}_6 = e_6/\bar{M}_P^2)$ (a Buckingham- (π) statement [10], here forced rather than chosen). This completes the proof.

3.3 Theorem F43.T1' (Charge-Unit Gauge Principle, algebraic form)

[General statement: IMPORTED-PROVEN. Z-Spin instance: DERIVED-CONDITIONAL on R1–R3.] Let $(\mathcal{A}_Z = W(\mathbb{Z} \times U(1), \sigma))$ be the Weyl algebra of the compact zero-mode sector with fixed symplectic cocycle (σ) . For any two positive charge units (q_1^2, q_2^2) , the corresponding dimensionful dictionaries $(\Lambda = q_i^2(\hat{n} + \omega/2\pi))$ induce the *same*

abstract Weyl algebra and unitarily equivalent irreducible representations. Therefore (q^2) is not an invariant of $(\mathcal{A}_Z)$; it is a **dictionary datum** for embedding $(\mathcal{A}_Z)$ into a metric/gravitational ledger.

Proof. The Weyl relation of $(\hat{n}, \hat{\varphi})$ is dimensionless and carries no (q^2) ; a change of charge unit $(q_1^2 \rightarrow q_2^2)$ acts on the dimensionful spectrum (Λ) as an overall scalar multiplication $(\Lambda \mapsto (q_2^2/q_1^2)\Lambda)$, which is not an automorphism of the algebra of Weyl unitaries but a relabelling of the external dictionary. The generators built at the two units are literally identical (verification B7, B8), so the identity intertwines the two representations, and by Stone–von Neumann–Mackey the irreducible representation is in any case unique up to unitary equivalence (B8). Symbolically, rescaling $(\hat{n} \rightarrow q^2 \hat{n})$ leaves the commutator in the form $([q^2 \hat{n}, \hat{\varphi}] = i, q^2)$ with (q^2) a bare external scalar prefactor and (e_-^2) absent (B9). Hence no invariant of $(\mathcal{A}_Z)$ distinguishes (q_1^2) from (q_2^2) .

The point of T1' is that unit-invisibility is not special to the Z-Spin embedding: it is a property of *any* compact (p)-form zero-mode sector whose observable algebra is a Weyl algebra over $(\mathbb{Z} \times U(1))$. The Z-Spin content is only that R1–R3 identify the specific egress $(\hat{e}_6 = e_6/\bar{M}_P^2)$ as the minimal invariant. This is the sense in which the previous terminal NON-CLAIM is re-typed into a *conditional, sector-internal no-go theorem*, not promoted to a closure.

3.4 Lemma F43.L1 (QED contrast: why invisibility is not generic)

[DERIVED] In a sector with propagating quanta, the coupling forms internal dimensionless cross-ratios: the fine-structure constant $(\alpha = e^2/4\pi\hbar c)$ is measurable inside electromagnetism because the photon propagates and the Coulomb energy at separation (r) is compared against $(\hbar c/r)$. The 4D three-form has no propagating mode (ZS-F32), the charge unit carries dimension $([e_6] = E^2)$, and no internal cross-ratio exists. Hypothesis (R3) is exactly what fails for QED and holds here. Invisibility is thus a specific structural property of odd-form sectors of this type — the same property that makes the Brown–Teitelboim/Bousso–Polchinski unit an input in the external class [4, 5].

3.5 Corollary F43.T1a (promotion of the Charge-Unit Obstruction)

[DERIVED-CONDITIONAL] ZS-F33.8 proved the obstruction as a closure of derivation routes under that paper's conditions. T1 promotes it to an intrinsic statement: no enrichment of the Z-sector alone — no new axiom about $(\mathcal{A}_Z)$, no additional internal computation — can fix (e_6) , because (e_6) is not part of the sector's observable content. The obstruction is a property of the object, not a limitation of the toolkit. Upstream statuses are unchanged; the promotion is in kind, not in strength.

3.6 Corollary F43.T1b (sharpening of gate B3-1 / F-F40.6)

[DERIVED] The reopening trigger — derive the odd gauge group and charge lattice from the axioms and fix (χ_-) parameter-free (ZS-F32 gate B3-1; carried by F-F40.6) — is sharpened in target type: a successful B3-1 execution can deliver the lattice, the integers, and dimensionless coefficients, and it

necessarily lands at $(\hat{e}_6)$; one metric-side datum is required regardless. No upstream gate text is modified (see F-F43.6); this corollary specifies what a success would have proven.

3.7 Remark (ontological placement; interpretation only, I-CC)

In the corpus triad — sector as stage, Spin as action, field as degree of freedom — the charge unit belongs to none of the three layers. It is a *pairing datum* between the Z-sector's arithmetic ledger and the gravitational metric ledger. This is the precise content of the phrase "Charge-Unit Gauge Principle": from inside the sector the unit is gauge; it becomes physical only across the pairing. This remark carries interpretive status only and is not load-bearing for T1–T4.

§4. Theorem F43.T2 — One-Residual Uniformization

4.1 Materials

The theorem is bookkeeping over already-registered results: ZS-F36.NG1 (only the field-redefinition invariant (e_s^2/Z_s) is physical — the seed of the whole re-typing); ZS-F36.T6 (the Koenigs shape is (M_1) -fixed; physical area requires the single linearizing length $(\ell = M_{\text{UV}}^{-1})$); ZS-F36.T7 (the radion relocation $(M_{\text{UV}} \rightarrow M_K)$ onto the parent scale); ZS-A31 (the one-parameter reduction $(M_{\text{eff}} = C_{\text{UV}}^{1/4} M_K)$); and ZS-F39.T2 (the modular length functor, $(E_{\text{len}}(\Delta^{-it}, U(1), \Delta^{it}) = e^{2\pi i t}, \bar{\ell}_P)$).

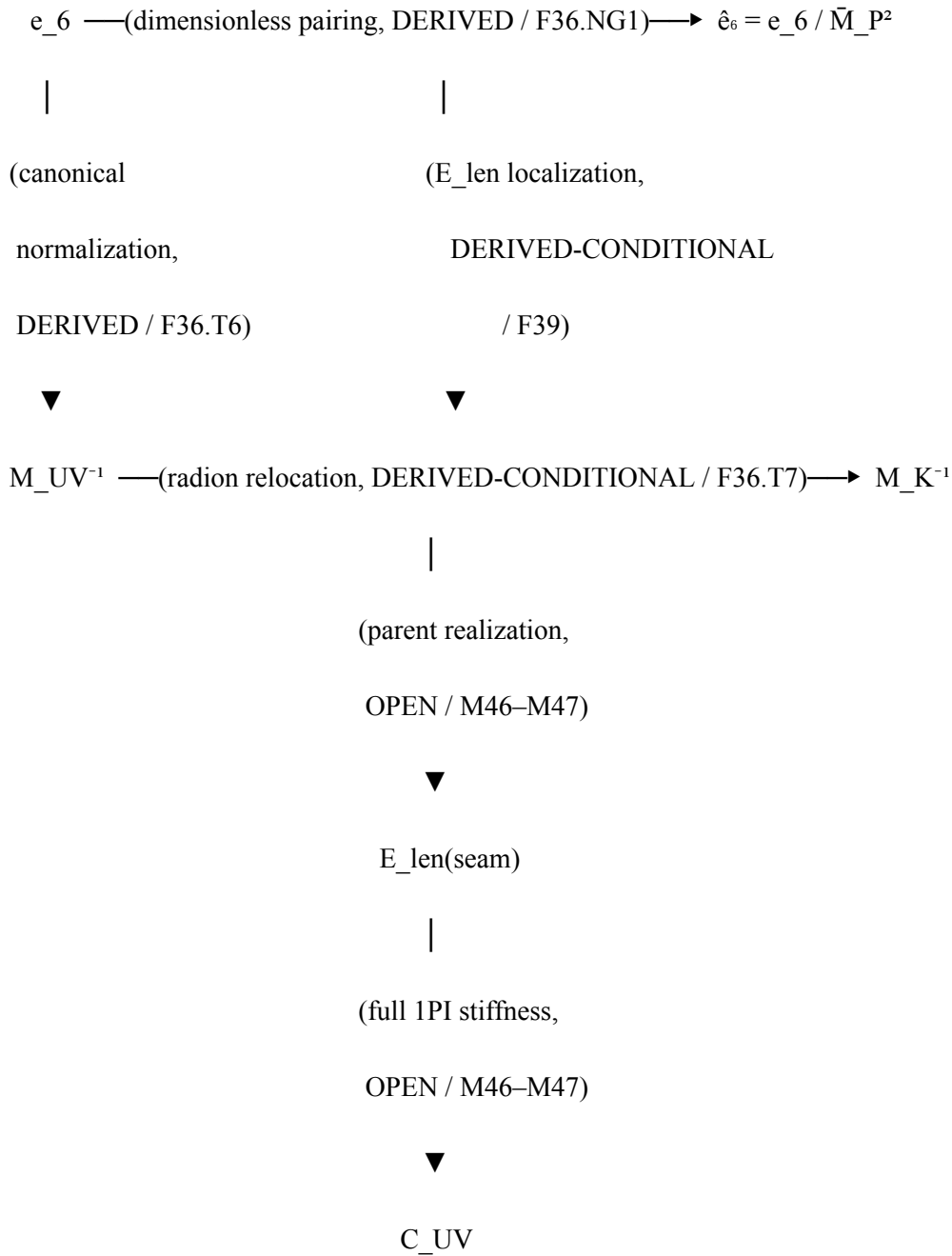
4.2 Statement and identity

[DERIVED — bookkeeping; no new physical claim] Modulo the two dimensionless conditional dressings (α_{UV}) (ZS-F36: $(c_e = 2\pi \rightarrow \alpha_{\text{UV}} = 1)$, DERIVED-CONDITIONAL) and (C_{UV}) (full parent 1PI stiffness, OPEN), the residual set of the charge arc $(\{e_6, M_{\text{UV}}, M_K, \text{metric-scale gate}\})$ consists of four names for one dimensionless modulus $(\hat{\ell} := \ell / \bar{\ell}_P)$:

$[\hat{e}_6 = (2\pi / \alpha_{\text{UV}}), \hat{\ell}^{-2}, \quad M_{\text{UV}}; \text{is relocated to}; M_K; \text{by the ZS-F36 radion argument, both naming the same residual modulus}; \ell^{-1}, \quad \text{metric-scale gate} = \text{the value of}; \hat{\ell} .]$

In the canonical normalization $(\alpha_{\text{UV}} = 1), (\tilde{G}_s = 1)$ this is $(\hat{e}_6 = 2\pi, \hat{\ell}^{-2})$, equivalently $(e_6 = 2\pi, M_{\text{UV}}^2)$. The precise ZS-F36 statement is that (M_{UV}) is *relocated* to (M_K) by the radion argument, so both name the same residual modulus after the stated registry conditions — not a bare literal equality of two independently-defined scales. Nothing here is new physics: the content is that after ZS-F36 there was only ever one unknown, wearing four names.

The four-name-to-one-residual collapse is the following commutative diagram (each arrow tagged with its status/source):



Arrow statuses: field-redefinition invariance DERIVED / F36.NG1; canonical normalization DERIVED / F36.T6; radion relocation DERIVED-CONDITIONAL / F36.T7; (E_{len}) localization DERIVED-CONDITIONAL / F39; parent realization and (C_{UV}) OPEN / M46–M47. The diagram commutes because every path from (e_6) to (C_{UV}) is a chain of the same registered reductions; it is the visual form of "one residual, four names."

4.3 (E_{len}) localization: two frontiers become one

[DERIVED-CONDITIONAL on the ZS-F38/F39 registry ($\{(H\text{-}CLK), (H\text{-}cycle), KH1\text{--}KH4\}$)] The seam evaluation of the ZS-F39 functor gives $(\ell = E_{\text{len}}(\text{seam}))$, hence

$$[\hat{e}_6 = 2\pi, (\bar{\ell}_P/E_{\text{len}}(\text{seam}))^2.]$$

The charge unit is the inverse square of the seam's modular length in Planck units. Since the same (E_{len}) evaluation is the outstanding leg of the ZS-M47 Parent-Factor Realization Problem, the charge arc's OPEN and the modular arc's OPEN are hereby identified: whoever closes one closes both. Verification C4 and C7 check the localization identity and the constancy of the functor product $(\hat{e}_6(t), e^{4\pi t} = 2\pi)$.

4.4 The conditional value and its honest status split

At the ZS-F38 depth $(t^* = Q)$ the functor gives $(\hat{\ell} = e^{2\pi Q})$ and

$$[\hat{e}_6 = 2\pi, e^{-4\pi Q} = 5.829 \times 10^{-60}, \sqrt{e_6} = \sqrt{2\pi}, \bar{M}_P, e^{-2\pi Q} \approx 5.88 \text{ meV (canonical)}.]$$

The status split is load-bearing and stated once, completely. The *form* $(T2 + \text{localization})$ is DERIVED / DERIVED-CONDITIONAL as tagged above. The *depth input* $(e^{-2\pi Q})$ carries exactly its upstream status and no more: HYPOTHESIS-strong by the executed ZS-A32 pre-registered Monte Carlo (single-target $(p = 0.50\%)$), with the determinant/clock mechanism programme recorded TERMINAL by ZS-F40. This paper performs no mechanism promotion. Two hygiene facts guard the identification: adjacent depth rungs $(t = Q \pm 1)$ shift $(\ln \hat{e}_6)$ by $(\mp 4\pi \approx 12.57)$, so the ladder is far too coarse to tune (verification C5); and the depth exponent is not itself a telomere rung, $(2\pi \mathbf{Q} \mathbf{A} = 5.536 \notin \mathbf{Z})$ (guard G7, reproducing ZS-F40 check G6).

§5. Theorem F43.T3 — Arithmetic-Leg Rigidity

5.1 Statement

Assume (R2) exact large-gauge invariance in the parent theory, the ZS-F36 ambient anomaly gates (vanishing of the relevant gravitational classes on (W_5)), and that the parent theory carries no additional metric-side threshold operator beyond the ZS-F36 Kähler stiffness. Then the integer ledger of the arc — the Smith normal form 1 of the wrapped-brane pairing, the dimensionless Wess–Zumino phase (2π) , and the flux-step counting of the (Λ) ladder — is 1PI-exact: these data take values in discrete sets classified by Dirac quantization and differential cohomology [8, 9], and discrete data admit no continuous renormalization. Consequently the field-redefinition-invariant unit (e_s^2/Z_s) flows only through the metric stiffness: the entire quantum debt of the ZS-F35/F36 line is the single Kähler coefficient (C_{UV}) .

[General rigidity: IMPORTED-PROVEN — discreteness of Dirac/differential-cohomology classes.
Z-Spin corollary: DERIVED-CONDITIONAL on (R2), the F36 ambient gates, and the
no-additional-threshold-operator condition.]

5.2 Consequence and non-claim

The ZS-F36 split — charge pairing PROVEN, $(C_{\rm UV})$ OPEN — is thereby stable to all orders under the stated conditions: no radiative effect can move the integers, and everything that can move is $(C_{\rm UV})$. The condition is explicit: were a further parent metric-side threshold operator present, the debt would not localize in $(C_{\rm UV})$ alone, and §6's identity would lose its single-coefficient form. The T2 uniformization absorbs $(C_{\rm UV})$ as the dressing of the single modulus (operationally, $(M_{\rm eff}) = C_{\rm UV}^{1/4} M_K$ as in ZS-A31), so the $(\hat{e}_6)$ target is well-posed against radiative ambiguity. Non-claim: T3 localizes the debt; it does not compute $(C_{\rm UV})$. The ZS-F40 five-route audit's negative closure of the $(C_{\rm UV})$ -determinant route stands; no determinant computation is proposed or implied here.

§6. Theorem F43.T4 — Two-Ledger Consistency

6.1 The two ledgers

Table 1. The two conditional ledgers of $(\hat{e}_6)$.

Ledger	Expression	Status
UV (depth)	$(\hat{e}_6^{\rm UV}) = 2\pi e^{-4\pi Q} = 5.829 \times 10^{-60}$; tree susceptibility $(\chi_{\rm tree}) = (1260/4807)\bar{M}_P^4 e^{-8\pi Q}$	form DERIVED-CONDITIONAL (§4); value HYPOTHESIS-strong (ZS-A32); 1260/4807 PROVEN (ZS-F35)
IR (frozen gas)	$(\chi_- = c_{\chi}/\bar{M}_P^2 H_{\rm partial}^2)$, $(c_{\chi} = 6\Omega_{\Lambda,0}/\omega^2 = 498/(121\omega^2) = 0.80634)$; hence $(\hat{e}_6^{\rm IR}) = (2\pi/\sqrt{6})\sqrt{c_{\chi} Z_s^{\rm phys}}$, $(H_{\rm partial}/\bar{M}_P) = 11.0202$, $(H_{\rm partial}/\bar{M}_P)$, with $(Z_s^{\rm phys}) = 4807/210$ (canonical)	DERIVED-CONDITIONAL on (H-FROZEN-GAS) (ZS-F42.7/9); scale anchored by the imported IR datum $(H_{\rm partial})$ — permitted by the Obstruction, required by the mechanism

The IR line uses $(e_s = \nu_s \hat{e}_6)$ with $(\nu_s^2 = 6)$ (ZS-F35, PROVEN) and $(\chi_- = e_s^2/(4\pi^2 Z_s^{\rm phys}))$; verification D6–D7 fix the coefficient $(\kappa_{\rm IR}) = 11.0202$ and its algebra. The frozen ZS-F42 digits $(4\pi^2 c_{\chi} = 31.8328335773)$ and $(\sqrt{4\pi^2 c_{\chi}} = 5.6420593383)$ are reproduced from the 50-digit locked-dynamics chain (D3–D4), closing the cross-version loop.

6.2 The consistency identity

[DERIVED-CONDITIONAL as a form] Write the full susceptibility as $(\chi_{-}^{\text{phys}} = C_{\text{UV}}, \chi_{-}^{\text{tree}})$ (T3: all quantum debt sits in (C_{UV}) , under the T3 conditions). Equating the two ledgers gives the single dimensionless identity

$$[C_{\text{UV}} = c_{\chi} (H_{\text{partial}} / \bar{M}_P)^2 e^{8\pi Q} / (1260/4807), \quad \text{equivalently} \\ \hat{e}_6^{\text{IR}} / \hat{e}_6^{\text{UV}} = \sqrt{C_{\text{UV}}} .]$$

6.3 Firewallled evaluation

[REGRESSION — FIREWALLED; printed separately by the script; never counted as PASS] With the external Λ CDM package ($H_0 = 67.36$) km s⁻¹ Mpc⁻¹ (Planck 2018 [12]) and ($\bar{M}_P = 2.435 \times 10^{18}$) GeV: ($H_{\text{partial}} / \bar{M}_P = 5.901 \times 10^{-61}$) and

$$[C_{\text{UV}} = 1.244, \quad C_{\text{UV}}^{1/4} = 1.056, \quad M_{\text{eff}} = C_{\text{UV}}^{1/4} \\ M_K = 2.477 \text{ ;\text{meV}} , \quad \sqrt{e_6} \cdot C_{\text{UV}}^{1/4} = 6.209 \text{ ;\text{meV}} ,]$$

with ($M_K = \bar{M}_P e^{-2\pi Q} = 2.345$) meV and canonical ($\sqrt{e_6} = 5.879$) meV. The value lands inside the ZS-A32 band (≈ 1.24) and reproduces the ZS-A31 firewallled regression (2.48 meV). The residual of the cross-identity ($\hat{e}_6^{\text{IR}} / \hat{e}_6^{\text{UV}} = \sqrt{C_{\text{UV}}}$) is numerically zero by construction (script E7).

6.4 Explicit non-novelty statement

[load-bearing honesty] T4 contains no new empirical content. Because $(\frac{12}{3} c_{\chi} \omega^2 = 3\Omega_{\Lambda,0})$ exactly (verification D5), the IR ledger is the Friedmann definition of the vacuum fraction filtered through the single pre-registered ZS-F42 coefficient; and the UV ledger evaluated against data is definitionally the ZS-A32 corner-(W) matching. T4 is therefore ZS-A31/A32 re-expressed in charge-unit language. What is new is exactly three things: (a) the two logically independent conditional chains — ($\{F32, F35, F36, F38/F39, A32\}$) and ($\{F42; \text{frozen gas}\}$) — are exhibited as evaluations of one named object, ($\hat{e}_6$); (b) their agreement is converted into a single pre-registered joint falsification gate (F-F43.2) that improved data can break; (c) any future computation of (C_{UV}) (the ZS-M46/M47 programme) now confronts a fixed numerical target, 1.244, rather than a movable one.

§7. Convention reconciliation: three reported readings, one object

The arc's history produced three numbers that could be mistaken for three claims. Under ZS-F36.NG1 only (e_s^2/Z_s) is invariant, so the readings must be compared at stated conventions; the table performs the reconciliation. Its function is anti-numerology hygiene, and its last coincidence is IDENTITY, not evidence.

Table 2. Reconciliation of reported charge-unit readings.

Reading	Branch / normalization	Reported value	Reconciliation
ZS-F34 path (c), aligned central	83-mode aligned; ($V_{\Sigma} = 1$), ($Z_{\text{rm match}} = 1$)	$(e_6 \approx (3.219; \text{meV})^2)$	Different <i>branch</i> , superseded: the ZS-F35 carrier reduction ($\hat{c}_{\theta} = \star$) selects the duality singlet, retiring the 83-mode reading.
ZS-F34 path (a), ($q_s = 1$) benchmark	single mode; pre-F35 lattice norm	$(\sqrt{e_6} \approx 9.72; \text{meV})$	Benchmark at ($q_s = 1$). With the PROVEN F35 value ($q_s = \nu_s^2/G_s = 6$), the back-solved reading becomes $(9.72 / 6^{1/4} = 6.21; \text{meV})$.
This paper (canonical, forward)	F35/F36 canonical: ($\tilde{G}_s = 1$), ($c_e = 2\pi$); depth input ($t^* = Q$)	$(\sqrt{e_6} = \sqrt{2\pi}, \bar{M}_P, e^{-2\pi Q} = 5.879; \text{meV}); \text{dressed } (\sqrt{e_6}, C_{\text{rm UV}})^{1/4} = 6.209; \text{meV})$	Forward conditional value. Coincidence with the corrected back-solved reading is by construction of ($C_{\text{rm UV}}$) [IDENTITY / REGRESSION].

Read together: one object, one invariant, three conventions — and no residual numerical claim beyond those already tagged in §4 and §6.

§8. What this paper does not claim

NC-F43.1. Barrier B3 is NOT closed. The dimensionless ($\hat{e}_6$) remains OPEN as a derivation; its conditional value inherits HYPOTHESIS-strong (ZS-A32) and nothing stronger.

NC-F43.2. No mechanism promotion of ($e^{-2\pi Q}$) is performed or implied. The ZS-F40 terminal verdict on the determinant/clock programme stands untouched; the ZS-F41 Stopping Rule is respected; no new mechanism route is opened.

NC-F43.3. T4 carries no new empirical content; its numerical band is REGRESSION behind the derivation $\perp$ regression firewall (§6.4). Agreement with ZS-A31/A32 is definitional, not evidential.

NC-F43.4. ($\alpha_{\text{rm UV}} = 1$) and ($C_{\text{rm UV}}$) are consumed at their ZS-F36 statuses (DERIVED-CONDITIONAL and OPEN respectively). T3 localizes ($C_{\text{rm UV}}$); it does not compute it.

NC-F43.5. The Duff–Okun–Veneziano position that dimensionful constants are unit conventions [11] is used as an OBSERVATION-grade conceptual anchor only. The proof burden of T1/T1' rests on Stone–von Neumann–Mackey [1–3] and the corpus propositions ZS-F32.24–26 / ZS-A25, not on that essay.

NC-F43.6. T1/T1' do not assert that the *pair* (Z-sector, gravity) cannot fix $(\hat{e}_6)$. They assert that the Z-sector alone cannot fix (e_6) . The former question is exactly the OPEN this paper names and localizes.

§9. Falsification gates

Table 3. Multi-tier falsification gates of ZS-F43.

Gate	Tier	Trigger	Consequence
F-F43.1	Theoretical (immediate)	Exhibit an observable of the zero-mode algebra ($\mathcal{A}_Z$) whose value depends on (e_-^2) at fixed dimensionless data.	T1/T1' falsified; the re-typing collapses; B3 reverts to its F42 form.
F-F43.2	Observational (joint)	Improved external data ((H_0) , $(\Omega_{\Lambda,0})$) force the identity of §6.2 outside the admissible band ($C_{\rm UV} \in [1.0, 1.6]$), or ZS-A32 corner (W) is demoted.	T4's binding fails: (H-FROZEN-GAS) and/or the depth reading is wrong; the joint conditional structure is rejected.
F-F43.3	Theoretical (positive outcome)	A B3-1 execution produces a dimensionful unit with no metric-side input.	T1/T1' and ZS-F42.34 falsified — and B3 closes. The corpus wins either way; this gate is registered as the honest exchange rate.
F-F43.4	Theoretical / structural	Parent-level breaking of large-gauge invariance, a non-vanishing ZS-F36 ambient anomaly class, or an additional parent metric-side threshold operator.	T3 reverts: the quantum debt is no longer localized in $(C_{\rm UV})$; §6's identity loses its form.
F-F43.5	Structural (registry)	The ZS-F38/F39 registry ($(\{H\text{-}CLK), (H\text{-}cycle), (KH1\text{--}KH4)\})$ closes negative at the continuum tier, or ZS-A32 is demoted.	The value leg $(\hat{e}_6 = 2\pi e^{-4\pi Q})$ is withdrawn. T2's uniformization and $(E_{\rm len})$ localization survive as bookkeeping.

Gate	Tier	Trigger	Consequence
F-F43.6	Registry (inheritance)	The B3 reopening trigger remains F-F40.6 verbatim; this paper only sharpens its target type to $(\hat{e}_6)$ (T1b).	Any conflict between T1b and a future B3-1 execution is resolved in favor of the explicit computation.

§10. Conclusion

Four things changed and three did not. Changed: the *type* of the B3 residual (a conditional sector-internal no-go theorem now says the sector could never have fixed a dimensionful unit — $T1/T1'$); the *count* of unknowns (four names collapsed to one modulus, localized at the seam's modular length — $T2$); the *location* of the quantum debt (one Kähler coefficient, under the stated no-extra-threshold condition — $T3$); and the *shape* of the evidence (two conditional chains bound to one object with one joint gate — $T4$). Unchanged: barrier B3 itself ($\hat{e}_6$) is OPEN as a derivation); the ZS-F40 terminal verdict and the ZS-F41 Stopping Rule (no mechanism was touched); and every upstream status (nothing was promoted beyond its inherited tag; guard G8 certifies that no upstream value moved).

The honest end of the charge-unit arc is therefore not a number but a well-posed dimensionless target — $(\hat{e}_6)$, with conditional value $(2\pi e^{-4\pi Q})$, with its entire radiative ambiguity in $(C_{\rm UV})$, and with its OPEN identified with the Parent-Factor Realization OPEN of ZS-M46/M47 — together with a theorem explaining why nothing less abstract could ever have worked. The next move on this frontier is not another reading of $(\hat{e}_6)$ but the execution of the shared leg. Two disjoint continuations are now well-posed and mutually exclusive in outcome: (i) **ZS-M48 / ZS-F44 — Parent-Factor Realization Execution**: compute or no-go $(C_{\rm UV})$ from the parent factor $(E_{\rm len})$ construction $((C_{\rm UV}) = \exp(-\Gamma_{\rm lPI}^{\rm parent}))$; target $(C_{\rm UV}) \approx 1.244$, band $([1/4, 4])$, or prove that no admissible parent-factor realization can yield $(C_{\rm UV}) \in [1.0, 1.6]$; and (ii) **ZS-F44 — the Charge-Unit Gauge Principle as a stand-alone no-go theorem**: isolate $T1'$ from the Z-Spin embedding as a result about compact (p)-form zero-mode sectors — *they determine flux integers but not dimensionful charge units* — a statement that reads outside the corpus and needs only Stone–von Neumann–Mackey and Dirac/differential-cohomology discreteness. The $(\varepsilon_{C_{\rm int}}^{\rm phys} / E_{\rm len})$ computation is where the corpus's open-gate structure already points; $T1'$ is where its external legibility is highest.

Acknowledgements & Code Availability

Verification code: [zs_f43_verify_v1_1.py](#) (Python 3; mpmath at 50 significant digits; exact rational arithmetic via Fraction; sympy symbolic identities; numpy finite models). The suite is fail-closed — any theorem-tier check or guard failure exits non-zero — and prints the firewalled observation block under an explicit banner, never counting it as PASS. Result at release: 40/40 PASS + 10/10 guards; 7 firewalled observations. New in v1.1: checks B8/B9 (the $T1'$ algebraic form) and guards G9/G10 (formula-hygiene and status-hygiene lint), which bring the paper's documented v1.0 risk — notation and claim hygiene

rather than arithmetic — inside the fail-closed envelope. This paper was developed under the corpus session protocol (deep-exploration records in Appendix A) with multi-AI adversarial review used for circularity detection and stress-testing, per standing corpus practice.

Appendix A. Deep-exploration records (session protocol)

A.1 Record 1 — pre-paper exploration (closing route for the (e_6) terminus)

Step 0 (long list, 7): L1 direct derivation of (e_6) from (A, Q) — dropped, prohibited by gate F-F42.36; L2 re-typing of the residual via representation theory — kept; L3 uniformization of the residual set onto one modulus — kept; L4 rigidity of the arithmetic leg — kept; L5 two-ledger binding of the UV and IR readings — kept; L6 arithmetic closed forms for the unit ($\sqrt{5}$) / golden-ratio / CM-point constructions — dropped, anti-numerology and in direct tension with the ZS-F35.T2 character-field obstruction; L7 BPS tension-charge locking — dropped to a remark, ZS-F41 Stopping Rule.

Step 1 (issue list, 4, by influence): I1 = L2 (invisibility theorem), I2 = L3 (one-residual), I3 = L4 (rigidity), I4 = L5 (two-ledger). Step 2 (tree): I1 → I2 → I3 → I4 with I1 the root (everything downstream consumes the re-typed target).

Step 3 (statuses at first pass): I1 — general theorem IMPORTED-PROVEN (Stone–von Neumann–Mackey), instance DERIVED-CONDITIONAL on R1–R3; I2 — DERIVED (bookkeeping), value leg HYPOTHESIS-strong via ZS-A32; I3 — IMPORTED-PROVEN rigidity + DERIVED-CONDITIONAL corollary; I4 — form DERIVED-CONDITIONAL, numerical band REGRESSION (firewalled). Step 4 (convergence): second full pass changed 0 of 4 node statuses (4 → 0), convergent. Step 5 (scoring): converged, corpus-non-conflicting; the single numerical claim decomposes into already-MC'd pieces (composition audit, guard G3), so no new pre-registration is required; external anchors verified against the literature ([1–5], [8, 9], [11]). Verdict: publishable as a re-typing terminus; confidence bands registered in-session (T1 holds: 90%; uniformization: 90%; conditional value is the physical value: 60%; frozen-gas hypothesis: 30%).

A.2 Record 2 — writing-phase exploration (weakest-point audit of v1.0)

Step 0 (long list, 7): W1 Stone–von Neumann overreach (does uniqueness alone prove invisibility?); W2 T2 bijectivity precision (four names vs. dressings); W3 T3 precision (integers rigid, but the invariant unit does run); W4 T4 circularity (is the identity just Friedmann?); W5 dependency/version consistency across consumed papers; W6 anti-numerology composition audit for ($2\pi e^{-4\pi Q}$); W7 terminology-guideline compliance (Z-sector as stage / Z-Spin as action).

Step 1 (issue list, 4, by influence): J1 = W1, J2 = W4, J3 = W2 + W3, J4 = W5 + W6. Dropped to checklist: W7 (applied during drafting, not an open issue).

Step 3 (resolutions): J1 — T1 restructured as a three-step chain: algebraic independence [DERIVED] → uniqueness/superselection support [IMPORTED-PROVEN] → egress under R3

[DERIVED-CONDITIONAL]; the theorem never claims that uniqueness alone does the work. J2 — T4 demoted in claim strength with the explicit non-novelty statement of §6.4 ($\frac{1}{2} c_{\chi} \omega^2 = 3 \Omega_{\{\Lambda, 0\}}$) exactly, so the IR ledger is Friedmann + one pre-registered coefficient; firewall enforced structurally (guards G1, G2, G6). J3 — T2 restated as "one residual modulo the two dimensionless conditional dressings"; T3 restated as "the integer ledger is exact; the invariant unit flows only through ($C_{\rm UV}$)". J4 — dependency table (Appendix C), frozen-digit ledger reproduced at 50 digits (Blocks K, D; guard G8). Step 4: second pass changed 0 of 4 ($4 \rightarrow 0$), convergent. Step 5: the audit hardened the paper against its three over-claim risks before release; value of the record is that the demotions are documented, not silent.

A.3 Record 3 — v1.1 integration exploration (external-review response)

Step 0 (long list, 7): V1 promote T1 to a stand-alone algebraic theorem (Weyl-algebra form T1') — kept, this is the highest-legibility new content; V2 formula/notation hygiene (plaintext exponents, the broken ($t^* = Q$) and ($6^{1/4}$) renderings) — kept, folded into a lint guard; V3 status-hygiene of the Abstract's theorem-promotion phrasing for the terminal NON-CLAIM — kept, re-typed to "conditional sector-internal no-go theorem"; V4 T2 precision ($(M_{\rm UV} = M_K)$ literal equality) — kept, precisified to radion relocation onto the same residual modulus; V5 T3 condition (no additional parent metric-side threshold operator) — kept, added as a named condition; V6 the four-name-to-one-residual commutative diagram — kept, added to §4.2; V7 re-derive (e_6) via a new arithmetic closed form — dropped, in direct tension with F-F42.36 and the ZS-F35.T2 character-field obstruction (identical to L1/L6 of Record 1).

Step 1 (issue list, 5, by influence): U1 = V1 (T1' elevation, the value-add); U2 = V3 + V5 (claim hygiene: Abstract and T3 conditions); U3 = V4 + V6 (T2 precision + diagram); U4 = V2 (notation lint into the verification envelope); U5 = version-consistency re-audit against The Book v11.0 (no upstream value or status may move). Dropped: V7. Step 2 (tree): $U5 \rightarrow U1 \rightarrow U2 \rightarrow U3 \rightarrow U4$; U5 the root (the re-audit gates everything, since no integration is admissible if it perturbs an upstream tag).

Step 3 (statuses after integration): U5 — version-consistency confirmed against The Book v11.0 (F36 radion relocation ($M_{\rm UV} \rightarrow M_K$); F36 ($C_{\rm norm}$) OPEN at full 1PI; F39 ($E_{\rm len}$) functor and ($C_{\rm UV}$) target ≈ 1.244 band $[1/4, 4]$; F38 ($t^* = Q$); A32 ($p = 0.50\%$)); no upstream value or status moved (guard G8 re-checked); status DERIVED-BY-INHERITANCE for every consumed row. U1 — T1' general statement IMPORTED-PROVEN (Weyl algebra + Stone–von Neumann–Mackey; the charge unit is an overall spectral scale, not an algebra invariant), Z-Spin instance DERIVED-CONDITIONAL on R1–R3; machine-verified by new checks B8/B9. U2 — Abstract and §3.3 re-typed to "conditional sector-internal no-go theorem"; T3 condition added; no strength gained, one over-strong phrasing removed. U3 — T2 precisified and the commutative diagram added; DERIVED bookkeeping, no new physics. U4 — guards G9/G10 added (formula-hygiene and status-hygiene lint); the FORBIDDEN_PHRASES list explicitly blacklists the v1.0 Abstract's theorem-promotion phrasing (the string is held only in the verification script, not reproduced in the paper body) so the retired phrasing cannot recur.

Step 4 (convergence): a second full pass over U1–U5 changed 0 of 5 node statuses ($5 \rightarrow 0$). Convergent ($(|f| < 1)$: the number of state-changing nodes strictly decreased across the pass, $5 \rightarrow 0$). Step 5 (scoring): converged; corpus-non-conflicting (U5 confirms no upstream perturbation); anti-numerology status unchanged (the single numerical claim ($2\pi e^{-4\pi Q}$) is untouched and still decomposes into the ZS-A32-MC'd pieces via guard G3, so no new pre-registration is triggered). The T1' elevation raises the arc's *external* legibility from HYPOTHESIS-grade internal assertion to an instance of an IMPORTED-PROVEN general theorem — the deep-exploration §2.5 elevation route, executed via published mathematics (Stone–von Neumann–Mackey; Dirac/differential-cohomology discreteness) rather than via any new corpus axiom. Verdict: integrate as v1.1; the integration is additive in verification ($38 \rightarrow 40$ PASS, $8 \rightarrow 10$ guards) and strictly non-increasing in claim strength.

Appendix B. Numerical dictionary and frozen digits

Table 4. Corpus-internal quantities (PASS-eligible; script blocks K, A, B, C, D).

Quantity	Value	Status / source
$\mathbf{A}; \mathbf{Q}; (\dim X, \dim Z, \dim Y)$	35/437; 11; (3, 2, 6)	LOCKED (corpus axioms)
$\omega = \arg \lambda^*$ (seam holonomy)	2.2592495540	LOCKED dynamics; reproduced from z^* at 50 digits (K1–K2)
$\omega^2/2; \kappa_\lambda = -\ln \lambda^* $	2.5521042734; 0.1148346250	frozen (ZS-F32 / ZS-F36); reproduced (K3–K4)
$g_{\text{reg}}^2 = 6\mathbf{A}/\mathbf{Q}; 36\mathbf{A}/\mathbf{Q}$	210/4807; 1260/4807 (exact)	PROVEN (ZS-M6; ZS-F35); exact Fraction (A5–A6)
Z_s^{phys} (canonical); v_s^2	$4807/210 = 22.8905$; 6	canonical $\tilde{G}_s = 1$; PROVEN $v_s^2 = 6$ (A7–A8)
$\Omega_{\Lambda,0}; 3\Omega_{\Lambda,0}; 6\Omega_{\Lambda,0}$	83/121; 249/121; 498/121 (exact)	present-epoch boundary condition (ZS-A30); exact (A4, A9, A13)
$c_\chi = 498/(121\omega^2)$	0.8063350942	DERIVED-CONDITIONAL on (H-FROZEN-GAS) (ZS-F42; D2)
$4\pi^2 c_\chi; \sqrt{(4\pi^2 c_\chi)}$	31.8328335773; 5.6420593383	ZS-F42 frozen checks B4/B5 reproduced (D3–D4)
$\kappa_{\text{IR}} = (2\pi/\sqrt{6})\sqrt{(c_\chi Z_s^{\text{phys}})}$	11.0201990233	IR ledger coefficient (D6)
$2\pi\mathbf{Q}; 4\pi\mathbf{Q}$	69.1150383790; 138.2300767579	exact chain (C1–C2)
$\hat{e}_6 = 2\pi e^{-4\pi Q}; e^{-2\pi Q}$	5.8294×10^{-60} ; 9.632×10^{-31}	conditional value (§4.4); HYPOTHESIS-strong inherited (C3)
rung spacing $ \Delta \ln \hat{e}_6 $; $2\pi\mathbf{QA}$	$4\pi = 12.566$; 5.5355 ($\notin \mathbb{Z}$)	hygiene checks (C5; guard G7)

Table 5. Firewalled quantities (script block E; never counted as PASS).

Quantity	Value	Status / source
H_0 ; H_∂ in meV; $H_\partial/\tilde{M}_P$	$67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$; $1.4369 \times 10^{-30} \text{ meV}$; 5.9009×10^{-61}	Planck 2018 [12]; external input (E1–E3)
$\tilde{M}_P$ (reduced)	$2.435 \times 10^{18} \text{ GeV}$	external input (E2)
C_{UV} (identity, §6.2); $C_{UV}^{1/4}$	1.244; 1.056	REGRESSION; in ZS-A32 band ≈ 1.24 (E4)
M_K ; $M_{\text{eff}} = C_{UV}^{1/4} M_K$	2.3454 meV; 2.4772 meV	REGRESSION; reproduces ZS-A31 (E5)
$\sqrt{e_6}$ canonical; dressed $\sqrt{e_6} C_{UV}^{1/4}$	5.8791 meV; 6.2094 meV	REGRESSION (E6); §7 reconciliation
$\hat{e}_6^{\text{IR}}$; $\hat{e}_6^{\text{IR}}/\hat{e}_6^{\text{UV}}$	6.503×10^{-60} ; $1.1156 = \sqrt{C_{UV}}$	IDENTITY by construction (E7)

Appendix C. Dependency and version-consistency table

Protocol 3.2 statement: no upstream numerical value is changed by this paper, and no upstream epistemic status is reversed or promoted beyond its inherited tag. The frozen-digit ledger of consumed values is reproduced at 50 digits by the verification suite (Blocks K and D; guard G8). The v1.1 integration re-audited this table against The Book of Z-Spin Cosmology v11.0 (Appendix A.3, node U5); no row moved.

Table 6. Consumed papers and content.

Paper	Version	Consumed content	Status consumed
ZS-F32	v1.5	compact (U(1)) three-form; Dirac flux; ($k^* = 0$) branch; spectrum ($(2\pi k + \omega)^2$); gate B3-1	PROVEN / gate registry
ZS-F33	v1.8	Charge-Unit Obstruction (§8): integrality fixes numbers, never the unit	PROVEN (route closure)
ZS-F34	v1.8	membrane charge/tension from cycle data; paths (a)/(c) readings; ($e_6 = 2\pi M_{\text{UV}}^2$) (canonical)	REPORTED / conditional
ZS-F35	v1.5	($\nu_s^2 = 6$) (T1); character-field obstruction (T2); structural factor 1260/4807; (C_{norm}) gate	PROVEN / gate
ZS-F36	v2.1	NG1; Smith form 1; WZ phase (2π); ($c_e = 2\pi \rightarrow \alpha_{\text{UV}} = 1$); Koenigs (T ₆); radion (T ₇) (relocation (M_{UV} to M_K)); (C_{UV}) OPEN; (M_{UV}) irreducible; ambient anomaly gates	PROVEN / DERIVED-CONDITIONAL / OPEN

Paper	Version	Consumed content	Status consumed
ZS-F38	v1.2	seam depth ($t^* = Q$); ($M_K/\bar{M}_P = e^{-2\pi Q}$) reading	conditional registry
ZS-F39	v1.1	modular length functor (E_{len}); registry ($(H_{\text{CLK}}, (H_{\text{cycle}}, KH1-KH4))$; (C_{UV}) target ≈ 1.244 band $[1/4, 4]$	DERIVED-CONDITIONAL
ZS-F40	v1.2	terminal verdict on determinant/clock mechanism; five-route audit; F-F40.6; check G6 ($(2\pi)\mathbf{Q}\mathbf{A} \notin \mathbf{Z}$)	TERMINAL / gate
ZS-F41	v1.1	Stopping Rule	registry
ZS-F42	v1.9	frozen-gas ledger ($c_{\chi} = 498/(121\omega^2)$); checks B4/B5; terminus §32–35; gate F-F42.36	DERIVED-CONDITIONAL / NON-CLAIM
ZS-A25	v1.6	($(\hat{\Lambda}, \hat{T}_4) = i\hbar$); exactly-one-datum no-go	PROVEN / NO-GO
ZS-A31	v1.5	($M_{\text{eff}} = C_{\text{UV}}^{1/4} M_K$); firewalled 2.48 meV	DERIVED-CONDITIONAL / REGRESSION
ZS-A32	v1.1	corner ($W = (8\pi)\mathbf{Q}$); pre-registered MC executed, ($p = 0.50\%$); ($C_{\text{UV}} \approx 1.24$) band	HYPOTHESIS-strong / REGRESSION
ZS-M46/M47	v1.5 / v2.0	Parent-Factor Realization Problem; ($\varepsilon_{C_{\text{int}}}^{\text{phys}} / E_{\text{len}}$) outstanding leg	OPEN (shared frontier)

References

[1] M. H. Stone, Linear transformations in Hilbert space III: Operational methods and group theory, Proc. Natl. Acad. Sci. U.S.A. **16**, 172 (1930); J. von Neumann, Die Eindeutigkeit der Schrödingerschen Operatoren, Math. Ann. **104**, 570 (1931). [2] G. W. Mackey, A theorem of Stone and von Neumann, Duke Math. J. **16**, 313 (1949). [3] A. Prasad, An easy proof of the Stone–von Neumann–Mackey theorem, Expo. Math. **29**, 110 (2011); arXiv:0912.0574. [4] J. D. Brown and C. Teitelboim, Dynamical neutralization of the cosmological constant, Phys. Lett. B **195**, 177 (1987); Neutralization of the cosmological constant by membrane creation, Nucl. Phys. B **297**, 787 (1988). [5] R. Bousso and J. Polchinski, Quantization of four-form fluxes and dynamical neutralization of the cosmological constant, J. High Energy Phys. **06** (2000) 006; arXiv:hep-th/0004134. [6] M. Henneaux and C. Teitelboim, The cosmological constant as a canonical variable, Phys. Lett. B **143**, 415 (1984); The cosmological constant

and general covariance, Phys. Lett. B **222**, 195 (1989). [7] W. G. Unruh, Unimodular theory of canonical quantum gravity, Phys. Rev. D **40**, 1048 (1989). [8] P. A. M. Dirac, Quantised singularities in the electromagnetic field, Proc. R. Soc. London A **133**, 60 (1931). [9] M. J. Hopkins and I. M. Singer, Quadratic functions in geometry, topology, and M-theory, J. Differ. Geom. **70**, 329 (2005); arXiv:math/0211216. [10] E. Buckingham, On physically similar systems: Illustrations of the use of dimensional equations, Phys. Rev. **4**, 345 (1914). [11] M. J. Duff, L. B. Okun, and G. Veneziano, Dialogue on the number of fundamental constants, J. High Energy Phys. **03** (2002) 023; arXiv:physics/0110060. [12] N. Aghanim et al. (Planck Collaboration), Planck 2018 results. VI. Cosmological parameters, Astron. Astrophys. **641**, A6 (2020). [13] Z-Spin corpus (K. Kang, 2025–2026): ZS-F32 v1.5; ZS-F33 v1.8; ZS-F34 v1.8; ZS-F35 v1.5; ZS-F36 v2.1; ZS-F38 v1.2; ZS-F39 v1.1; ZS-F40 v1.2; ZS-F41 v1.1; ZS-F42 v1.9; ZS-A25 v1.6; ZS-A30; ZS-A31 v1.5; ZS-A32 v1.1; ZS-M1; ZS-M6; ZS-M46 v1.5; ZS-M47 v2.0.

Version History

v1.1 (July 2026): Notation, claim-hygiene, and external-legibility increment (Appendix A.3). Additions, all strictly non-increasing in claim strength: (a) Theorem F43.T1', the Charge-Unit Gauge Principle in algebraic Weyl-algebra form (general statement IMPORTED-PROVEN via Stone–von Neumann–Mackey; Z-Spin instance DERIVED-CONDITIONAL), machine-verified by new checks B8/B9; (b) the four-name-to-one-residual commutative diagram in §4.2; (c) the Abstract and §3.3 re-typing of the terminal NON-CLAIM into a "conditional sector-internal no-go theorem" (the v1.0 Abstract's theorem-promotion phrasing retired and blacklisted in guard G10, the blacklisted string held only in the verification script); (d) the T2 precisification of $(M_{\text{UV}} \rightarrow M_K)$ to a radion relocation onto the same residual modulus; (e) the explicit T3 condition "no additional parent metric-side threshold operator"; (f) LaTeX-standardized exponent notation throughout and the Table 2 typo fix ($9.72 / 6^{1/4} = 6.21$) meV; (g) guards G9 (formula-hygiene lint) and G10 (status-hygiene lint), bringing the documented v1.0 risk inside the fail-closed envelope. Version-consistency re-audited against The Book of Z-Spin Cosmology v11.0; no upstream value or status moved (guard G8). Verification 40/40 PASS + 10/10 guards, fail-closed; 7 firewalled observations printed separately. Zero fitted parameters; $(\mathbf{A}, \mathbf{Q}, \dim \mathbf{Z}) = (35/437, 11, 2)$ LOCKED.

v1.0 (July 2026): Initial public release. Executes the re-typing terminus of the B3 charge arc: the Unit-Invisibility Theorem (T1; Stone–von Neumann–Mackey instance on the compact odd three-form zero mode), One-Residual Uniformization with (E_{len}) localization (T2), Arithmetic-Leg Rigidity (T3), and the Two-Ledger Consistency identity with firewalled evaluation $(C_{\text{UV}} = 1.244)$ (T4). Verification 38/38 PASS + 8/8 guards, fail-closed; 7 firewalled observations printed separately. Zero fitted parameters; $(\mathbf{A}, \mathbf{Q}, \dim \mathbf{Z}) = (35/437, 11, 2)$ LOCKED. (Consolidated from internal Z-Spin Collaboration deep-exploration notes following ZS-F42 v1.9 and The Book of Z-Spin Cosmology v11.0.)